



**CANDIDATE
NAME**

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CLASS

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**INDEX
NUMBER**

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BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

23RD AUGUST 2017
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Section A

Answer **all** questions in the spaces provided on the Question paper.

Section B

Answer **one** question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
SECTION A	
1 [30]	
2 [12]	
3 [8]	
SECTION B	
4 [25]	
OR	
5 [25]	
TOTAL P3 [35%]	75

This document consists of **21** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** questions in this section.

- 1 Milk is an important source of diet for most infant. For many adults, milk is an important source of dietary calcium. Milk contains many biological molecules; one of them is lactose. Therefore, it is not surprising that most infant will have a mechanism to digest lactose. While for adults, it is not normal for their body to be able to digest lactose. However, there is an increasing trend of lactose tolerant (also known as lactase persistent) individual in the adult population. Here, we will discuss 3 types of conditions; namely Congenital Lactase Deficiency (CLD), lactose allergy, and lactose tolerance (lactase persistence).

Silanikove et al The Interrelationships between Lactose Intolerance and the Modern Dairy Industry: Global Perspectives in Evolutional and Historical Backgrounds Nutrients 2015, 7, 7312-7331

- (a) Fig. 1.1 shows the structure of lactose.

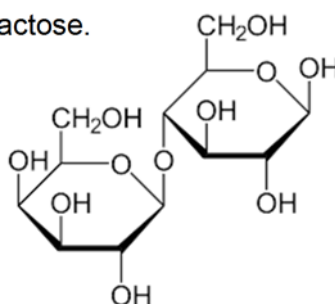


Fig. 1.1

Name the bond joining the 2 monomers in lactose

ANS [L1] (novel)

1. **β -1,4-glycosidic bond**

[1]

- (b) Fig. 1.2 shows the catalytic residues found in the active site of lactase.

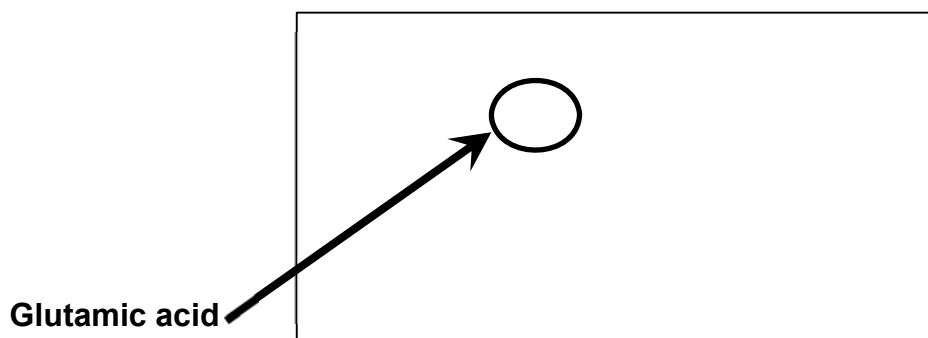


Fig. 1.2

One of the catalytic residues; glutamic acid (circled in Fig. 1.2) is substituted by glycine which is shown in Fig.1.3 below.

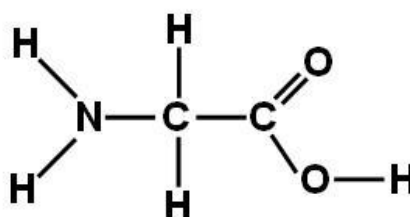


Fig. 1.3

Explain how lactase catalytic activity is affected by the substitution above.

ANS [L3] (novel)

[2]

1. **Glutamic acid** has an **R-group** that is **negatively charged** whereas **glycine** has an **R-group** that is **non-polar**.
2. This causes the **change in the interaction** between the **catalytic residues** and the **substrate** at the **active site**; therefore; lactase catalytic activity will be **greatly reduced / lost**.

(c) Lactose intolerance in infant is also known as Congenital Lactase Deficiency (CLD). It is an autosomal recessive disorder.

Studies have shown that CLD is caused by mutation in *LCT* gene coding for lactase. The most commonly observed mutation is a single nucleotide substitution in *LCT* gene which results in the production of truncated lactase.

Other mutation such as a single nucleotide deletion in *LCT* gene has also been detected in a few patients which also results in the production of truncated lactase.

(c)(i) Explain how two different types of mutations; single nucleotide substitution and single nucleotide deletion in *LCT* gene can lead to the production of truncated lactase.

ANS [L3] (novel)

[2]

1. **Single nucleotide substitution** may cause the codon to become a **stop codon**, resulting in a **nonsense mutation** which leads to the production of truncated lactase.
2. **Single nucleotide deletion** may cause frameshift mutation, resulting in an early encounter of stop codon downstream the deletion which leads to the production of truncated lactase.

(c)(ii) A small amount of DNA is isolated from infants suffering CLD resulted from a single nucleotide substitution. The DNA was subjected to process Y to ensure enough DNA for the subsequent Southern Blotting process.

Name the process Y.

ANS [L1] (novel)

[1]

1. **Polymerase chain reaction (PCR)**

Fig. 1.4 shows wild type *LCT* gene and mutant *LCT* gene. The mutation result in the loss of *NdeI* restriction site (*NdeI* RE) at position 4 kb as shown by the arrow .

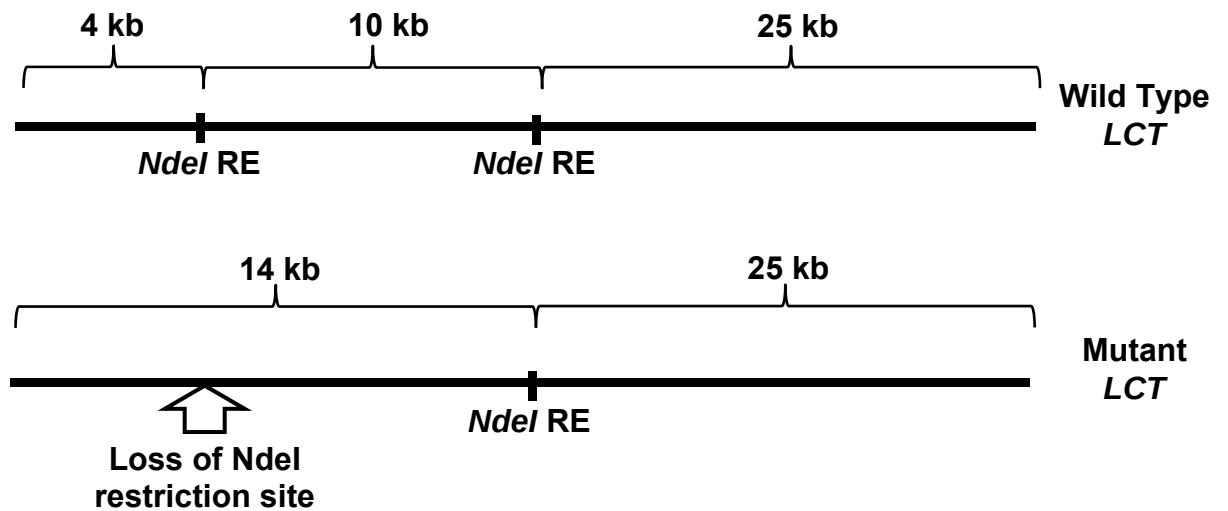


Fig. 1.4

Fig. 1.5 shows the band patterns of the nitrocellulose membrane obtained from the Southern Blotting of DNA sample from 3 infants.

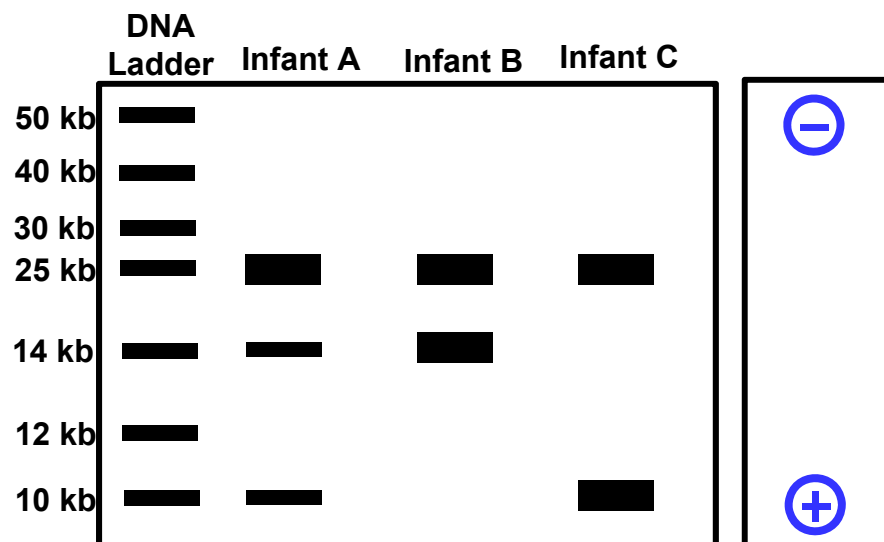


Fig. 1.5

(c)(iii) On the box on the right side of Fig. 1.5, indicate the position of the positive terminal and negative terminal during the gel electrophoresis which results in the band pattern in Fig.

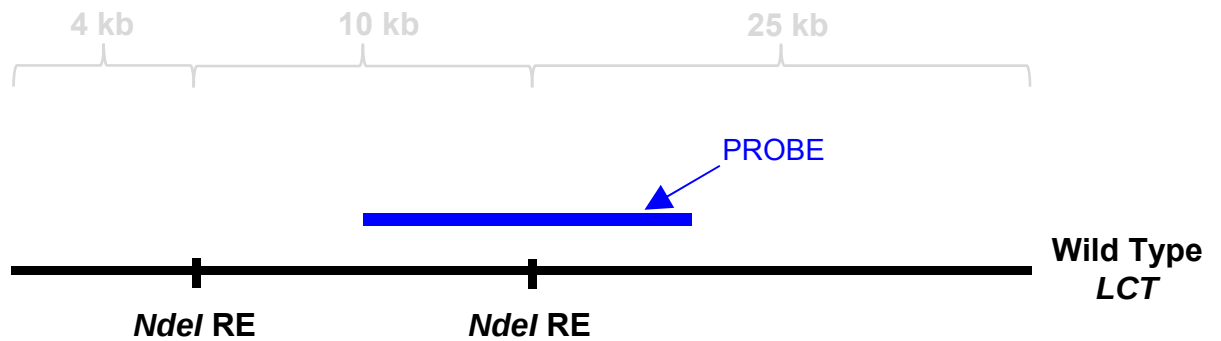
1.5.

[1]

ANS [L2] (novel)

[1]

- (c)(iv) Based on the position of the *NdeI* restriction sites in wild type and mutant *LCT* gene in Fig. 1.4 as well as the band patterns in Fig.1.5, indicate on the wild type *LCT* below where the probe will anneal to (use a ruled line and label). [1]



ANS [L3] (novel) [1]

- (c)(v) Based on the information provided in part (c) as well as Fig. 1.4 and Fig. 1.5; explain which infant is suffering from CLD. [2]

ANS [L3] (novel) [2]

1. Infant B is an infant with CLD as infant B is **homozygous for mutant LCT** / carries **2 copies of mutant LCT**.
2. When digested with *NdeI*, the LCT gene produced **14 kb and 25 kb** fragments only / **double band thickness at 14 kb and 25 kb**.

- (d) Another condition known as lactose allergy results in more severe symptoms than lactose intolerance. The symptoms of allergy are due to the action of Immunoglobulin E (IgE) which activates mast cells, which subsequently secrete a chemical signal X.

- (d)(i) Name X.

ANS [L1] (Novel)

1. **histamine**

[1]

Fig. 1.6 shows the structure of IgG and IgE.

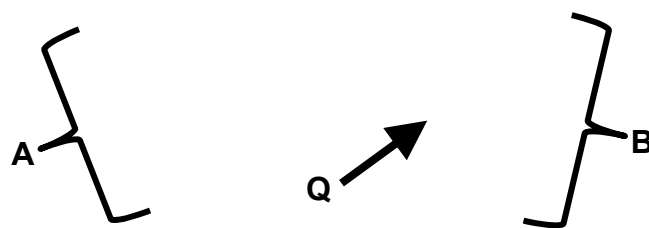


Fig. 1.6

- (d)(ii) The number, type, and position of Q on IgG and IgE are different. Name precisely the process which attaches Q on IgG and IgE.

ANS [L1] (Novel)

[1]

1. glycosylation

- (d)(iii) Describe structures of IgE that allow it to perform its role in eliciting allergy response towards lactose.

ANS [L2] (Novel)

[2]

1. The **variable region** of IgE is **complementary to lactose** allowing it to **recognise lactose**.
2. The **constant region** of IgE is **able to interact with mast cell** to **activate it** and elicit allergy response towards lactose.

- (d)(iv) With reference to Fig. 1.6; suggest what will happen to an individual with lactose allergy when part **B** of all his IgE is replaced with part **A** of his IgG

ANS [L3] (Novel)

[1]

1. IgE will still be **able to bind to lactose** to **elicit response** that is usually **elicited by IgG**.

Some human adults continue to produce the lactase enzyme throughout their adulthood (lactase persistent). Therefore, they are able to digest lactose effectively (lactose tolerant) and will not develop symptoms such as bloating, flatulence, or diarrhoea after consuming milk.

However, most adult mammals stop producing the lactase enzyme (lactase non-persistent). Therefore, they are unable to digest lactose effectively (lactose intolerant) and will usually develop symptoms such as bloating, flatulence, or diarrhoea when consuming milk.

Fig. 1.7 shows the pattern of inheritance of lactose intolerance in Family A.

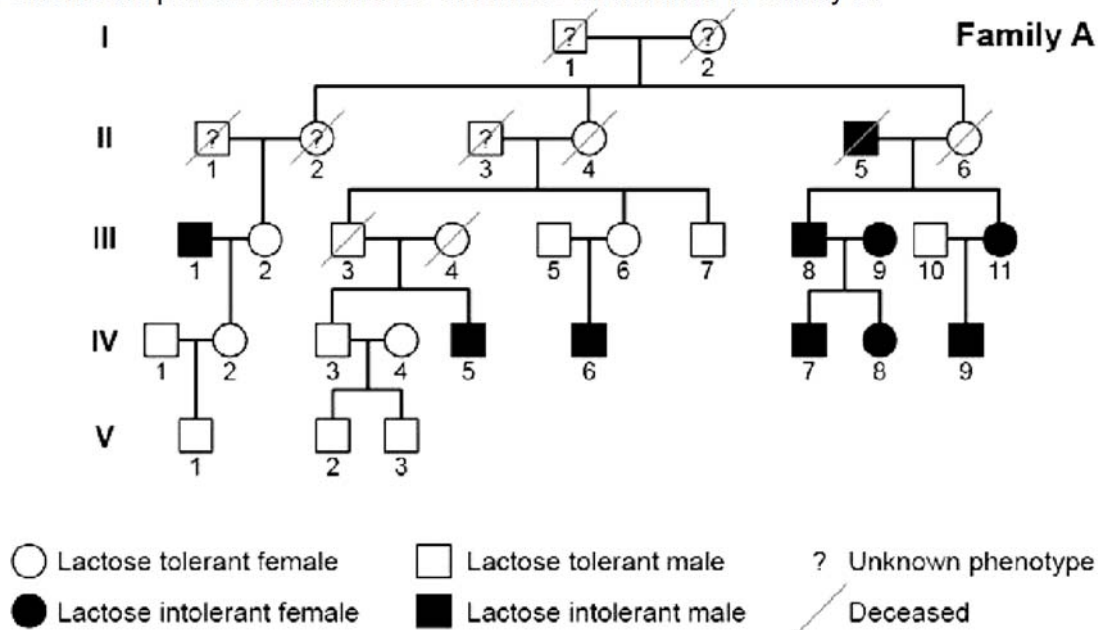


Fig. 1.7

- (e)(i) With reference to Fig. 1.7; explain the mode of inheritance of lactose intolerance and where the gene is probably located. Provide evidence to support your claim.

ANS [L2] (novel)

[4]

1. Lactose tolerance is recessive
2. GIII5 x GIII6 [both Lact tolerant] >>GIV6 Lact Intolerant
3. Location of gene: Autosomal

4. Given that **both male and females inherit equally**.

Humans learned to exploit ruminants as a source of milk about 10,000 years ago; particularly the European population. Since then, the use of domesticated ruminants as a source of milk and dairy products has expanded until today when the dairy industry has become one of the largest sectors in the modern food industry, including the spread at the present time to countries such as China and Japan.

Widespread lactose intolerance among the adult population is a considerable drawback to dairy-based foods consumption. Over the centuries, three factors allowed humans to overcome limitations imposed by lactose intolerance: (i) mutations, which occurred in particular populations, most notably in the north European Celtic societies and African nomads, in which carriers of the lactose intolerance gene converted from being lactose intolerant to lactose tolerant; (ii) the ability to develop low-lactose products such as cheese and yogurt; and (iii) colon microbiome adaptation, which allow lactose intolerant individuals to overcome its intolerance.

Fig. 1.8 shows the pockets of lactase persistence shown in pie charts.



Pockets of lactase persistence shown in pie charts. Ingram et al, Hum Genet 124:579–591, 2009.

Fig. 1.8

(e)(ii) With reference to Fig. 1.8; apart from the genetic factors, suggest what other factor could have contributed to the spread of lactase persistence.

ANS [L3] (novel)

[2]

1. genetic trait influenced by **Cultural factors**
2. There is a relationship between the **frequency of lactase deficiency** in a population and **whether or not the population was involved in intensive dairy farming**. OWOTTE
3. Low levels of lactase deficiency are found in European populations with a long history of dairy farming, and highest levels in populations of Asian /American ancestry who were not dairy farmers.

(f)(i) Explain how Darwin's principles of evolution may be applied in understanding the type of evolution that would have had to take place in the spread of lactose tolerance.

ANS [L2] (Novel)

[3]

1. **Natural selection** is the first principle in Darwin's Evolution that would be seen taking place, where the **selection pressure** is the **availability of dairy produce**.
2. **Variation** in the **gene pool** aided through **mutation** that allowed the continuation of the gene lactase ie **lactase persistence**, instead of natural loss of function [lactase non-persistent];

3. Allowing these **advantageous trait [lactase persistence] to be selected for** and allow individuals to **survive** better as they were now able to **exploit the additional resource** to their advantage, and **reproduce** so that the **Advantageous trait is passed** on to the offspring and the next generation ;
4. **divergent evolution** took place and over time allowed the evolution of lactose tolerant individuals ;

(f)(ii) Suggest how the scenario described in (f)(i) is an example of macro or micro evolution.

ANS [L3] (Novel)

[2]

1. **microevolution;**
2. distinct changes in **allele frequency** within the population making it more **distinct**, allowing lactase persistence the advantage over lactase non-persistence;

(g)(i) Define anthropomorphic climate change.

ANS [L1] (Novel)

[1]

1. refers to **the production of greenhouse gases** emitted by **human activity**.

(g)(ii) Explain how the above scenario of increased lactase persistence may contribute to anthropomorphic climate change.

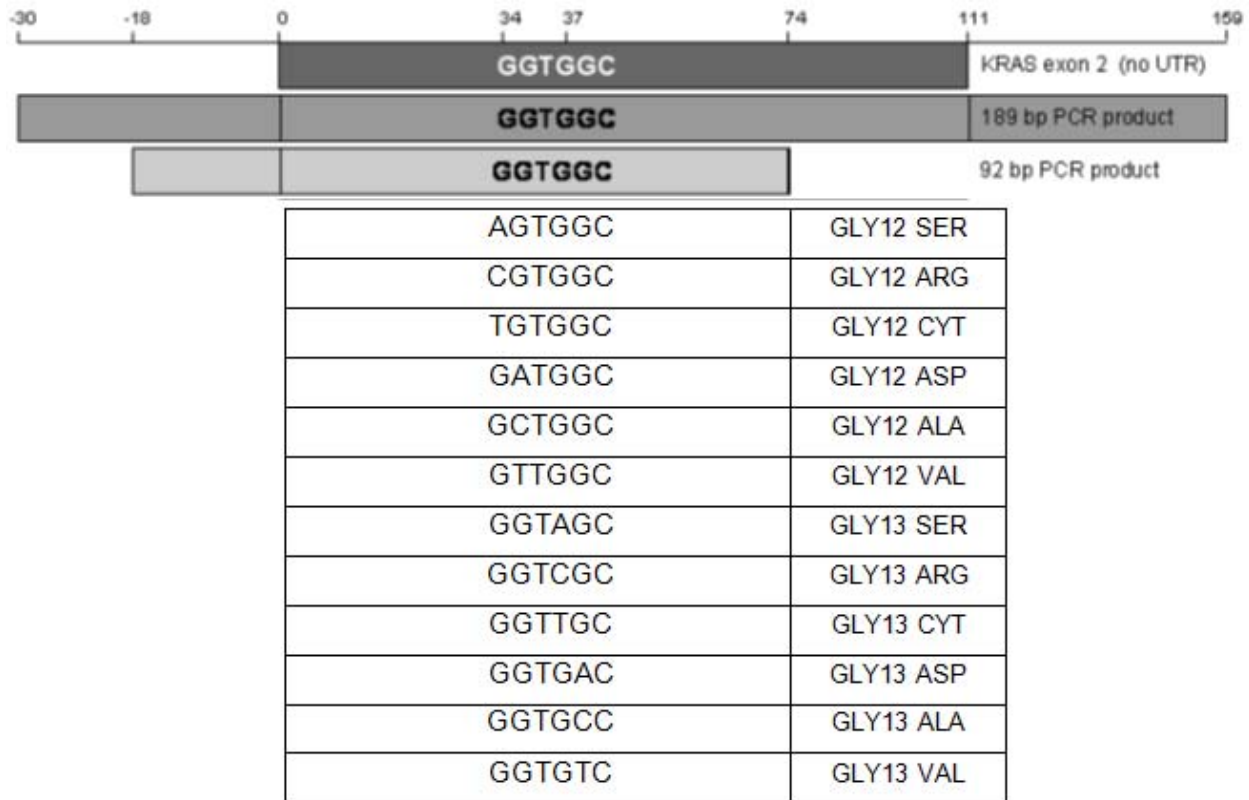
ANS [L2] (Novel)

[3]

1. Lactase persistence **confers and evolutionary advantage** in being able to **exploit a wider range of food especially dairy products**. This **indirectly contributes** to anthropomorphic climate change in terms of the **human choice of food**, increased **demand of dairy products**.
2. Impact [max 1]
 - 2a] **deforestation** – **arable land** is created for **crop feed and raising of cattle**, there is the **loss of terrestrial carbon sink**
 - 2b] **cattle industry** – increased production of **methane and CO₂**
 - 2c] **Dairy industry** – use of **fossil fuel** in the processing of dairy products e.g. **pasteurization** / and as a result **releasing more CO₂ / carbon foot print larger**.
3. **Green house gas emissions** especially **CO₂ [Green house Warming Potential (GWP):1] and Methane GWP 24** contribute to **global warming and impact climate change**.

[Total: 30]

- 2 All mammalian cells express three closely related Ras proteins: H-Ras, K-Ras and N-Ras that promote oncogenesis when mutationally activated at codons 12, 13 or 61. Despite a high degree of similarity between the isoforms, K-Ras mutations are far more frequently observed in cancer.



Location of K-RAS codon 12 and 13 mutations and PCR amplicons / products. Exon 2 of K-RAS is shown from the ATG without the untranslated region. The position and size of the PCR amplicons used in the High Resolution Melting HRM assays in relation to exon 2 of K-RAS is indicated. All possible mutations at codon 12 and 13 are listed along with the corresponding amino acid changes from Glycine (GLY) are shown.

Krypup et al. High resolution melting analysis for the rapid and sensitive detection of mutations in clinical samples: KRAS codon 12 and 13 mutations in non-small cell lung cancer BMC Cancer 2006, 6:295

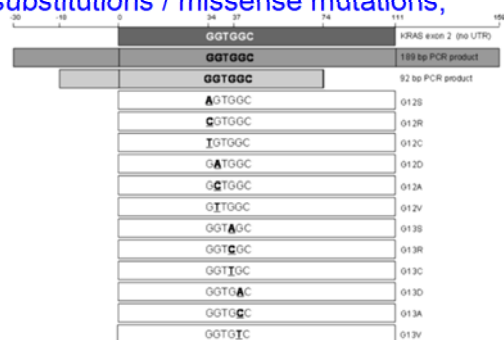
Fig. 2.1

(a) With reference to Fig. 2.1; explain the type of mutation experienced in K-RAS codon 13. [2]

ANS [L2] (Novel)

- all mutations found on codon 13 are base pair substitutions / missense mutations, N37 GGC to N37 AGC;
- resulting in a change of one amino acid, from GLY to SER;

NB: Minus one mark for no example cited



(b) With reference to Fig.2.1; suggest why oncogenesis in codon 13 is caused by only 6 possible changes in amino acid and not more.

ANS [L3] (Novel)

- due to base pair substitutions on the 1st and 2nd bases of codon 13, subsequently causing change in one amino acid;
- base pair substitution on 3rd base of codon 13 would have a silent mutation, i.e. not result in any change in amino acid ;
- due to the genetic code being degenerate;
- deletion or addition point mutations result in frame shift and subsequent loss of function of RAS which would not result in cancer. / stunted cell division.

Ras proteins are the products of proto-oncogenes that are frequently mutated in human cancers. They are encoded by three ubiquitously expressed genes: *H-Ras*, *K-Ras* and *N-Ras*. These proteins are GTPases that function as molecular switches regulating pathways responsible for proliferation and cell survival.

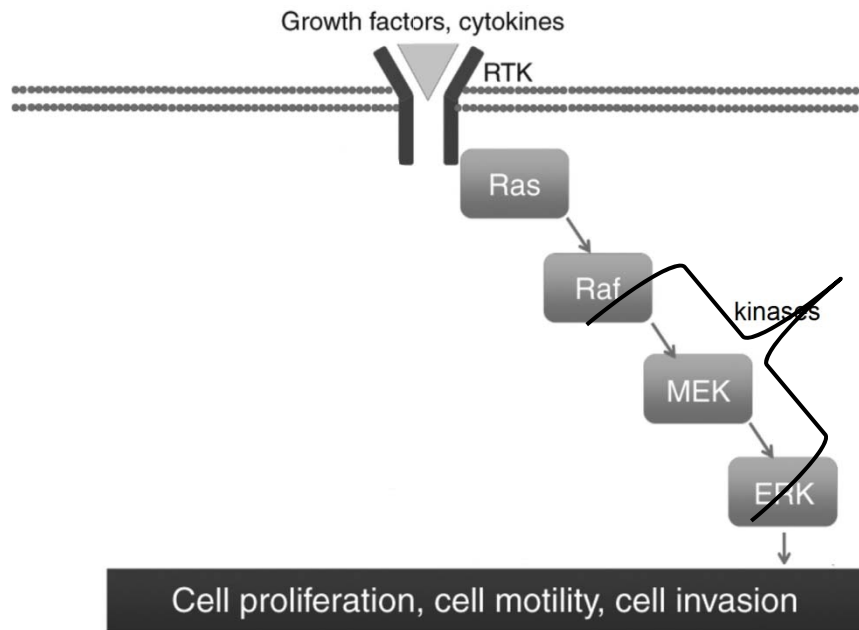


Fig. 2.2

(c) How does Ras protein act as a molecular switch in initiating cell proliferation?

ANS [L2] (Novel)

[1]

1. RAS is activated by Tyrosine kinase which includes a GTP within RAS / replacement of GDP with GTP, which turns the switch on activating protein kinases Raf, MEK and ERK.

(d) With reference to Fig. 2.2; explain how Ras protein is normally used to terminate cell proliferation.

ANS [L2] (Novel)

[2]

1. When **no growth factors / cytokines is bound to RTK**, Ras acts as a **GTPase** which causes the **hydrolysis of GTP to GDP** therefore inactivating RAS and
2. **Phosphorylation cascade** involving Raf, MEK, and ERK cannot continue and phosphatases are recruited to remove the phosphate; inactivating the protein kinases which terminate cell proliferation.

(e) With reference to Fig. 2.2 suggest how a mutation in *Ras* gene results in uncontrolled cell division.

ANS [L3] (Novel)

[1]

1. **Gain of function mutation in Ras gene** results in **Ras protein** which is **constitutively active / hyperactive**; requiring **no activation by RTK that is bound by growth factors / cytokines**. As a result the signaling pathway involving Raf, MEK, and ERK that lead to cell proliferation is constitutively activated; OR
2. Mutation in RAS would result in a **gain of function** which results in **GTP remaining** within RAS due to a loss of its GTPase function, allowing RAS to **remain turned on / active even after the growth factors / cytokines has been removed from RTK**. As a result the signaling pathway involving Raf, MEK, and ERK that lead to cell proliferation is constitutively activated.

(f) With reference to (a) to (e); explain the development of cancer.

ANS [L3] (Novel)

[2]

1. Development of cancer as a multistep process
2. requiring the mutual activation of codons 12, 13 and 61. OWTTE
3. requiring the accumulation of mutations in these codons
4. acquiring a gain of function such that system is independent of the growth hormone

[Total: 12]

- 3 Cyanobacteria are a group of bacteria that obtains their energy through photosynthesis. They carry an operon known as *phycocyanin* operon which controls the expression of phycocyanin. Phycocyanin is a protein complex which serves as accessory pigment to chlorophyll in cyanobacteria. Without phycocyanin, light harvesting process is halted. The amount of phycocyanin increases from very low level to high level in the presence of light.

Fig. 3.1 below shows the structure of *phycocyanin* operon in the Cyanobacterium *Anacystis nidulans*.

P	O	<i>CPCB1</i>	<i>CPCA1</i>	Intergenic region	P	O	<i>CPCB2</i>	<i>CPCA2</i>
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Legend:

P	:	promoter
O	:	operator
<i>CPCB1</i> and <i>CPCA1</i>	:	structural genes coding for β – subunit of phycocyanin
<i>CPCB2</i> and <i>CPCA2</i>	:	structural genes coding for α – subunit of phycocyanin

Fig. 3.1

- (a) Compare the *lac* operon to the *phycocyanin* operon.

ANS [L3] (Novel)

[3]

Features	<i>lac</i> operon	<i>phycocyanin</i> operon
Similarity:		
Type of operon	Inducible operon	Inducible operon
Differences:		
Number of promoter	1 promoter controlling the expression of 3 structural genes; <i>lacZ</i> , <i>lacY</i> , and <i>lacA</i>	2 promoters controlling the expression of 2 structural genes for β -subunit of phycocyanin and 2 structural genes for α -subunit of phycocyanin.
Inducer	Allolactose	Light

- (b) Describe how the *CPCA2* gene could be transferred to another bacterium by a prophage.

ANS [L2] (Novel)

[3]

- When the prophage viral DNA is excised from the chromosome, it takes with it ***CPCA2* gene** located **adjacent bacterial DNA** due to **improper excision. OWTTE**
- CPCA2* gene** is **injected**, along with the phage's genome, into the next host cell. This DNA remains **double-stranded during transfer**.
- Both strands** are **integrated** and may subsequently **replace** the **homologous region** found on the **chromosome** of its **new host**.

- (c) Cyanobacterium *Anacystis nidulans* has thylakoid which contain the same photosystems as the thylakoid of plant cells. Fig. 3.2 shows a hybrid *phycocyanin* operon.

trp P	O	CPCB1	CPCA1	Intergenic region	trp P	O	CPCB2	CPCA2
-------	---	-------	-------	-------------------	-------	---	-------	-------

Legend:

trp P : trp promoter

Fig. 3.2

Explain the rate of production of oxygen in the Cyanobacterium *Anacystis nidulans* carrying the hybrid operon when light is present and tryptophan is present.

ANS [L3] (Novel)

SC: Explain
OR: Describe with reason

rate of production of oxygen
very low or negligible

hybrid operon
controlled by trp P

light, trp present
trp co-repressor
the operon is off

[2]

- The rate of production of oxygen is **very low or negligible**; the hybrid operon is under the control of **trp promoter** which is **off when tryptophan is present**.
- No expression of phycocyanin; photoactivation halts**. Therefore, **photolysis cannot occur to produce oxygen**.

[Total: 8]

Section B

Answer **one** question in this section

Write your answers on separate answer paper provided.
Answer each part on a **separate** piece of paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4** (a) Explain the fluid mosaic model and the roles of the constituent biomolecules in functions of membranes at the cell surface and of membranes within the cell. [13]
- (b) Explain how genetic variation arises in a natural population and its significance in allowing the population to adapt and evolve. [12]

[Total: 25]

- 5** (a) With reference to named examples, describe the roles of proteins in bringing about cell signalling in living organisms. [13]
- (b) Gene expression in eukaryotes is regulated at many different stages of the process. Explain how gene expression is regulated in eukaryotes and the significance of this at each stage. [12]

- 4 (a) Explain the fluid mosaic model and the roles of the constituent biomolecules in functions of membranes at the cell surface and of membranes within the cell. [13]

ANS [L3] Novel

[13]

FLUID MOSAIC MODEL (Max 2)

1. **Fluid** - 'Fluid' refers to the fact that individual phospholipid and protein molecule is able to **diffuse laterally** within the bilayers. It also refers to phospholipids being able to **diffuse transversely** across the bilayers.
2. **Mosaic** - 'Mosaic' describes the **assorted pattern** produced by the **scattered integral and peripheral protein molecules** which differ between each bilayer.

PHOSPHOLIPIDS (Max 4)

3. Arrange themselves in a layer 2 molecules thick **bilayer**;
4. 4a] Arrangement confers **stability**;
4b] where the polar hydrophilic phosphate head interacting with aq. medium in/s and o/s cell;
4c] non-polar hydrophobic fatty acid tails help enclose a non-polar hydrophobic interior away from the surrounding aqueous medium;
5. 5a] Acts as a **control for movement of substances into and out of cell** as the hydrophobic core restricts movement of ions, large polar molecules;
5b] forcing these molecules to enter via selective channels conferring selectivity to membrane;
5c] act as a barrier to separate the content of the cell from the external cell environment.
6. **Regulate membrane fluidity**, hence **permeability** through **proportion of the saturated fatty acids to unsaturated fatty acids**. Saturated fatty acids are closely packed together, causing the membrane to be rigid; Unsaturated fatty acids have spaces between them due to the double bonds and this increases membrane fluidity.
7. **Also regulate membrane fluidity through length of the fatty acids**. Fluidity increases due to the presence of more short fatty acid chains.

CHOLESTEROL (Max 3)

8. **regulates fluidity** by **regulating the movement of phospholipids**. Prevents phospholipid bilayer from becoming too fluid or too rigid and also help **maintain membrane stability**.
9. **decrease fluidity** by **partially immobilising unsaturated fatty acid tails**;
10. **increase fluidity** by **spacing out saturated fatty acid chains**;

11. **low temperatures** ⚡ cholesterol disturbs the close packing of phospholipids & keeps them **more fluid / increasing permeability**
12. **high temperatures** ⚡ cholesterol prevents membranes from breaking up and makes membrane **less fluid** by restraining the movement of phospholipids / **decreasing permeability**

MEMBRANE PROTEINS (Max 4)

13. **Include Peripheral and Integral proteins;**
14. **Function in Transport.** Transmembrane protein that spans the membrane provides a **hydrophilic channel** across the membrane that is selective for a particular solute. E.g. aquaporins (water channels) found on the collecting duct on the kidney nephron.
15. Other **transport proteins** hydrolyze ATP as an energy source to **actively pump** substances across the membrane. E.g. sodium-potassium **pump** on the neurone that utilizes ATP to pump 2 K⁺ ions in for every 3 Na⁺ ions out.
16. Function as **Enzymes**. A protein built into the membrane may be an enzyme with its active site exposed to the substances in the adjacent solution needed to carry out sequential steps of a metabolic pathway. E.g. mitochondrial ATP synthase.
17. Function in **Cell Signalling**. A membrane protein that has a binding site complementary in shape to the chemical messenger which upon binding causes a conformational change in the protein that relays message to the inside of the cell. E.g. insulin receptors on the hepatocytes of the liver.
18. Function in **Intercellular joining**. Membrane proteins of adjacent cells may be hooked together in various kinds of junctions. E.g. Tight junctions on the proximal convoluted tubule of the nephron.
19. Function in **Cell to cell recognition**. Glycoproteins (proteins with short chains of sugar) that serve as identification tags and are recognized by other cells. E.g. antigens on erythrocytes that confer specificity.
20. Function in **Attachment to the cytoskeleton and the extracellular matrix**. Attachment of microfilaments or other elements of the cytoskeleton to help maintain cell shape and fix the location of certain membrane proteins. These proteins can coordinate extracellular and intracellular changes.

CARBOHYDRATES, GLYCOLIPIDS AND GLYCOPROTEINS (Max 2)

21. Oligosaccharides that are covalently bonded to lipids formed **glycolipids**; Oligosaccharides that are covalently bonded to proteins formed **glycoproteins**.
22. As **recognition sites** for **cell-to-cell recognition** (cell identity markers); **receptor sites for chemical signals**
23. For **cell-to-cell adhesion** ⚡ Helps cells maintain structural relationships with neighbouring cells
24. **As Antigens for RBC / MHC or surface Ag for immune response;**

QWC: Scientific argumentation exemplified by:

Two or more examples of constituent biomolecules of membranes linked coherently functions of membranes at the cell surface and of membranes within the cell

- 4 (b) Explain how genetic variation arises in a natural population and its significance in allowing the population to adapt and evolve. [12]

ANS [L2] Novel

[12]

Causes of genetic variation in a population. (Max 8)

Gene Reshuffling (Max 2)

1. Gene reshuffling is the rearrangement of genes and alleles to produce a **new combination of alleles** within the gene pool of a population.
2. **Independent Assortment** during **Metaphase I** allows for **new combination of alleles** to be formed within a gamete. The possible number of combinations of gametes produced by one human would then be $2^{23} = 8388608$ (8 million) combinations.
3. **Crossing over during Prophase I** between **non-sister chromatids of homologous chromosomes** allows for a **recombinant chromatid** to be formed, **forming new combinations of alleles within a gamete**.
4. **Random Fertilization of gametes** allows for a **new combination of alleles within a zygote**.

Gene Mutation (Max 2)

5. Gene mutation is a form of mutation where there is a **change in the nucleotide sequence or nucleotide number** of the alleles of a gene. This includes **Substitution, Duplication, Insertion/addition, Deletion and Inversion**.
6. Gene mutation **gives rise to new alleles** and **increases the gene pool** of a population.

Chromosomal Structural Aberration (Max 2)

7. Structural aberration includes **Deletion, Duplication, Inversion and Translocation**. **Structural aberration gives rise to new alleles** and increases the gene pool of a population.
8. **Structural aberration** occurs **during crossing over in meiosis**. In crossing over, non-sister chromatids sometimes **exchange unequal-sized segments of DNA**, so that one partner gives up more genes than it receives. The products of such a **nonreciprocal crossover** are one chromosome with a **deletion** and one chromosome with a **duplication**.

Chromosomal Numerical Aberration (Max 2)

9. **Numerical aberration** refers to the change in the **number** of chromosomes, leading to either **aneuploidy or polyploidy**.
10. Numerical aberration occurs when there is a **non-disjunction** during meiosis, **where members of a pair of homologous chromosomes do not move apart properly** during meiosis I or **sister chromatids fail to separate** during meiosis II.

Environmental Influence (Max 2)

11. Common environmental factors (Light, Diet and Temperature) may influence the **degree of phenotypic expression** or in some cases, change the phenotype entirely.
12. 1 example of environmental factor influencing phenotype cited

significance in allowing the population to adapt and evolve by natural selection (Max 6)

13. **Natural Selection** is the mechanism which **acts on individual organisms in a population** and those with **favourable phenotypes** which are morphologically, physiologically and behaviourally **better adapted** to the **prevailing selection pressure** within the **existing environment** are at a **selective advantage** and hence are more likely to **survive to reproductive age to produce viable offspring** / greater reproductive success and in doing so **pass down their favourable alleles**.
14. Those individuals with **unfavourable phenotypes** that are **not so well adapted** are at a **selective disadvantage** and either **fail to reproduce or die before they can reproduce**. Over **many generations**, the **proportion of the favourable alleles INCREASES** in the population. This then leads to **ADAPTIVE EVOLUTIONARY CHANGE**.
15. Genetic variation within a population **provides the raw material on which natural selection works** (i.e. variation is a **pre-requisite for evolution by natural selection**)
16. When **environmental changes** occur, variations allow some individuals with certain **favourable characteristics to survive better** and **reproduce more successfully** than others, to produce fertile offspring.
17. If there is **no variation** in the population, all will be **equally susceptible** to the effects of environmental change and it is possible that the entire population will be **wiped out**. Variation thus helps to **ensure perpetuation** of species and **safeguard species from extinction**.
18. Citing example to illustrate significance of variation in wild varieties or related organisms being useful in agriculture (e.g. *Teosintes*, evolutionary cousins of corn, carry genes for resistance to diseases affecting domesticated corn and have been used to produce disease-resistant corn varieties.)

QWC: Scientific argumentation exemplified by:

Two or more examples illustrating how genetic variation can from sexual reproduction / meiosis and genetic mutation linked coherently how these variations are important in allowing the population to adapt and evolve by natural selection that to the correct stages of the process

- 5 (a) With reference to named examples, describe the roles of proteins in bringing about cell signalling in living organisms. [13]

ANS [L2] (H2 JC2 CJC MYE/2011/P2/Q8(a) Modified) [13]

Each stage of cell signalling must be clearly stated.

Stage: Ligand-Receptor Interaction (Max 5)

Cell Surface Receptor Proteins

1. Certain **cell surface membrane proteins** function as **specific receptor proteins** with **binding sites complementary in conformation (shape)** to particular chemical signal molecules (called **ligands**) - **ligand-receptor interaction** (binding of ligand to receptor) **initiates a signal transduction pathway** INSIDE the cell

G-Protein Linked Receptor (GPLR)

2. Binding of extracellular **ligand (signal) molecule** such as **glucagon, adrenaline** to the **binding site** of the **G-Protein Linked Receptor (GPLR)** cause it to undergo a **conformational change**.
3. **Activated GPLR** binds to the **G_α subunit** of inactive G-protein, inducing a **conformational change** and cause **attached GDP** to be displaced (from the G_α subunit) and **replaced by a GTP molecule; G_α subunit dissociates from the G_{βγ} subunit**.
4. Activated **G protein** (or activated G_α subunit) then **binds** with and activate other **effector membrane proteins within the cell**.

GTPase Enzyme

5. Present as **part of the G-protein** that is associated with **GPLR (G-protein-linked receptors) or GPCR (G-protein-coupled receptors)**;
6. Once specific cellular responses have been carried out, **GTP** attached to an activated G protein is rapidly **hydrolysed** to **GDP** by the intrinsic **GTPase** enzyme (in the **G_α subunit**, causing G_α subunit to dissociate from the effector membrane protein and reassociate with the G_{βγ} subunit). This **inactivates** the **G-protein** and **switches off** the **function of the activated G-protein** / ref. to role of the G-protein in signal transduction;
7. Allows whole **system can be shut down quickly** when the extracellular signal molecule is no longer present.

Adenylyl Cyclase

8. **Found** in the **plasma membrane**; in close association with G-proteins;

9. **Activated by** the binding of **activated G-proteins**; G-proteins are activated as a result of the binding of hormones / ligands (e.g. adrenaline / glucagon) to **G-protein linked receptors**;
10. Upon activation, catalyses **ATP to cAMP** which acts as a second messenger, (diffuses through the cell and) activates protein kinase A (a serine/threonine kinase), which phosphorylates other proteins;

Tyrosine-Kinase Receptors

11. Present as **sections of the 2 TKR (tyrosine-kinase receptor) polypeptides**;
12. Enzymes **activated** by the **binding of ligands** to both of the receptor polypeptides and subsequent **dimerization**;
13. Activated tyrosine kinase on one polypeptide **adds phosphates** to the **tyrosine tails** of the **other polypeptide**;
14. The fully-activated receptor proteins activate a variety of specific relay proteins that bind to specific phosphorylated tyrosine molecules.

Phospholipase C.

15. **Activated** when a **signal molecule binds** to the G-protein linked membrane receptors or tyrosine kinase receptors;
16. Upon activation, **cleaves** a membrane **phospholipid (PIP₂)**, into 2 by-products - **diacylglycerol (DAG)** and **inositol trisphosphate (IP₃)**;
17. the IP₃ acts as a second messenger to activate a gated-calcium channel, releasing Ca²⁺ from the cell's endoplasmic reticulum, thereby increasing the cytosolic Ca²⁺ concentration;

Ion-channel Receptors

18. Transmembranal ligand-gated **ion channel proteins** with **hydrophilic channels** and an **extracellular ligand-binding site**.
19. The hydrophilic channels / pores **open or close** (due conformational change in channel protein) in response to **binding by ligand molecules** such as acetylcholine to **regulate** (i.e.to allow or prevent) the **passage of specific ions**, e.g. Ca²⁺, **into** or **out** of the cell

Stage: Signal Transduction (incl. phosphorylation and signal amplification) (Max 5)

Protein Kinase A / Protein Kinase

20. **Activated by** second messenger, **cyclic AMP (cAMP)**;
21. Upon activation, brings about **phosphorylation of other inactive protein kinases** (ref. to mostly the serine or threonine amino acids of these protein kinases being affected), activating them;
22. The **subsequent phosphorylation** of other inactive protein kinases by these **activated protein kinases** leads to a '**phosphorylation cascade**' which brings about a widespread cellular mechanism for regulating protein activity;
23. Each protein **phosphorylation changes the shape of the protein kinase** (due to the interaction between the phosphate group and charged or polar amino acids) that typically **converts it from an inactive form to an active form** which in turn activates the subsequent kinase;

Protein Phosphatases

24. Remove **phosphate groups** from **activated protein kinases**, **inactivating them**;
25. causing the **signaling pathway** and the **subsequent cellular response to shut down**;
26. responsible for **turning off a signal-transduction pathway** in the absence of the extracellular signal molecules;

Stage: Cellular Response (Max 5)

Intracellular Receptors

27. Ref. to intracellular receptors that interact with **ligand molecules** that are **steroidal in nature** (e.g. sex hormones) and able to **dissolve through the cell surface membrane**
28. **Activated protein receptors** (in the form of **hormone-receptor complex**) in turn act as **transcription factors** that control which genes are **turned on** and are **transcribed** into messenger RNA (mRNA) in cellular response
29. Ref. to proteins such as transcription factors, RNA polymerase to bring about transcription
30. Ref. to proteins such as ribosomal proteins as part of ribosomes to bring about translation

- 5 (b) Gene expression in eukaryotes is regulated at many different stages of the process. Explain how gene expression is regulated in eukaryotes and the significance of this at each stage. [12]

ANS [L3] (modified from specimen paper /P3/Q5b) [12]

Regulation of gene expression / protein synthesis – at various stages

Any **seven** from below:

1. Chromatin Level / Chromatin modification – by Methylation of DNA
2. Histone modification of DNA
3. Specific Transcription Factors / Repressors & Activators
4. Control Elements, incl. promoters, silencer / enhancer, DNA sequences
5. Post-transcriptional control / pre-mRNA processing
6. RNA / intron splicing / polyadenylation / 5' capping
7. Translational Control
8. Half-life of mRNA / initiation of translation
9. Post-Translational control
10. Biochemical modification to make functional protein, including protein degradation
11. AVP

Advantages of regulating gene expression

Any **four** from:

12. Chromatin level
 - idea of longer term switching genes on or off to restrict active genes to cells that required (by the cell line), do more efficient / less waste of resources Reject: inactivation / imprinting
13. Transcriptional Level
 - allows rate of production to be regulated to, match short term requirements / allows flexibility
14. Post-transcriptional level
 - allows for production of different variants / regulates stability of process
15. Translational level
 - idea of affecting how long the process takes to stop
16. Post-translational level
 - allows rapid production of product from stored precursor / product can be activated where it is needed / allowing, safe transport / storage of, inactive form / ref. to phosphorylation for immediate responsiveness to cell conditions / AVP
17. AVP

QWC: Scientific argumentation exemplified by:

Two or more advantages of regulating gene expression / protein synthesis linked coherently to the correct stages of the process